Industrial Visual Inspection: From Prototype to Operational Decision

Executive summary · independent project on public benchmark images (VisA), not factory data

Technical feasibility on public data only. A separate engineering review recomputed the reported results from the raw score files and found them consistent; the project owner's review and acceptance are pending.

Problem and prototype

On a production line, defects are rare and varied, so labelled defect examples are scarce. The question: can a model that has only seen good boards send the right circuit-board images to a human reviewer, without burying the reviewer in false alarms?

The prototype uses frozen pretrained CNN features and a per-position Gaussian (Mahalanobis) model of normal appearance, registers each board, and flags an image for review when its score exceeds a threshold frozen beforehand on separate normal images. It is fitted on normal images only — by design; VisA's defect labels and masks are used only for evaluation. It recommends a review and never rejects a board.

GO / NO-GO

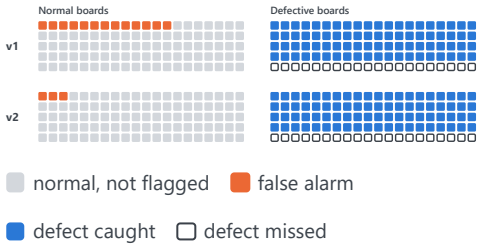
No-Go for a manufacturing pilot. Although v2 improved false-positive performance, it did not meet the proposed detection-sensitivity and robustness requirements. It is therefore not recommended for a manufacturing pilot without further work. Rule-based verdict against the unchanged criteria: INCONCLUSIVE.

Go for one bounded validation step — no further model work before it.

Controlled improvement on PCB2, a board type not used in development

v1 → v2: one pre-registered change — score only inside the board region. 100 defective + 100 normal test images, thresholds frozen before testing, evaluated once; 95% confidence intervals.

	v1	v2	Change, v2 − v1
Defect recall	80%	**80%**	unchanged
False alarms, test normals	13%	**3%**	−10 pts (CI −17 to −3)
False alarms, holdout normals	5.3%	**4.0%**	−1.3 pts (CI −4.6 to +2.0)
AUROC	0.927	**0.979**	+0.052 (CI +0.028 to +0.082)



Fewer false alarms and better ranking; recall did not improve and stays below the proposed 90% target. Median latency per image (GPU): v1 194 ms, v2 224 ms.

Follow-up finding on PCB3, a separate board type

A pre-registered global brightness/contrast correction met its own decision rule ("ROBUSTNESS IMPROVEMENT SUPPORTED"; darkening difference −4.7 pts, CI −8.6 to −0.8), but that result is **weakly informative**, and the correction made a mild blur raise false alarms by +14.5 pts. **Decision: do not adopt it.** The uncorrected pipeline is not production-ready either: brightening raised its false alarms by +15.2 pts. PCB3 numbers are not comparable with PCB2's.

Disclosure: after PCB3 test scoring, one input check in the analysis program was repaired (7dbd302); no score, threshold or decision changed. Full record in the repository.

Remaining risks

- **Detection:** recall 80% against a proposed 90%; the smallest third of defects only 56%.
- **Lighting:** a 10% darker image raised PCB2 false alarms from 3.6% to 28.7%.
- **Portability:** on a green background the original board-finding rule marked 96% of the frame as board, silently.
- **Evidence base:** benchmark images and synthetic perturbations; targets are the developer's proposal. No end user, factory image, pilot or financial return.

Next recommended validation step

- Agree requirements with a quality engineer: cost of a missed defect versus one review, review capacity, defects that matter.
- Run the unchanged tool in shadow mode on real line images, affecting no decision.
- Entry gates: board segmentation and registration on the site's own normal images; a per-image lighting check; a labelled sample of real defects, with recall reported by defect size.
- Decide Go / No-Go again on that evidence.

Data: VisA dataset, Zou et al., ECCV 2022, CC BY 4.0 (github.com/amazon-science/spot-diff). Pretrained network: torchvision Wide-ResNet-50-2 (ImageNet weights). PCB1–PCB3 are benchmark board types, not manufacturing sites. Not affiliated with or endorsed by the dataset authors or any company.

Built with AI coding assistance (Claude Code) under the project owner's direction; owner review pending. Project owner: Leon Lai. Protocols, raw results and the two-page extended brief: project repository.